

Exercise Sheet 3

Deadline for submission: Wednesday, 15 October 2025 at 08:00.

Discussed in exercise session: Friday, 17 October 2025 at 10:15.

Exercise 3.1 - Markov Matrices: Coffee in Central America

We are given some rare information about the trading network of coffee within cities of Costa Rica and Panama. More specifically, we know how much the individual cities trade with each other each month. Given an initial distribution of the total coffee in the cities, **find out which city will ultimately have the most coffee.**

- (a) Complete the code in the provided Jupyter notebook `pagerank.empty.ipynb`.
- (b) What assumptions do we make about the coffee trade for this result to be valid? Is that a realistic setting?

Exercise 3.2 - Machine Epsilon

Use the provided Jupyter notebook `MachineEpsilon.empty.ipynb` for this exercise section. Compute the sum

$$s := 1 + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \cdots - \frac{1}{10^7}$$

in two different ways in Python:

- a) From left to right.
- b) From right to left.
- c) Compare the outcomes to the exact solution

$$s = 1.19314713055994780942\dots$$

and explain the background of the different outcomes. Why do we not get the same results?

- d) Write a code to determine the machine epsilon from your computer experimentally by using a `while`-loop.

Exercise 3.3 - Iterative Improvement

Assume we are given a linear system of the form $A\mathbf{x} = \mathbf{b}$ and an approximate but inexact solution $\mathbf{b}_{inexact}$. With the method of Iterative Improvement it is possible to iteratively improve the provided inexact solution.

Complete the required function in the Jupyter notebook `iterative_improvement.empty.ipynb`